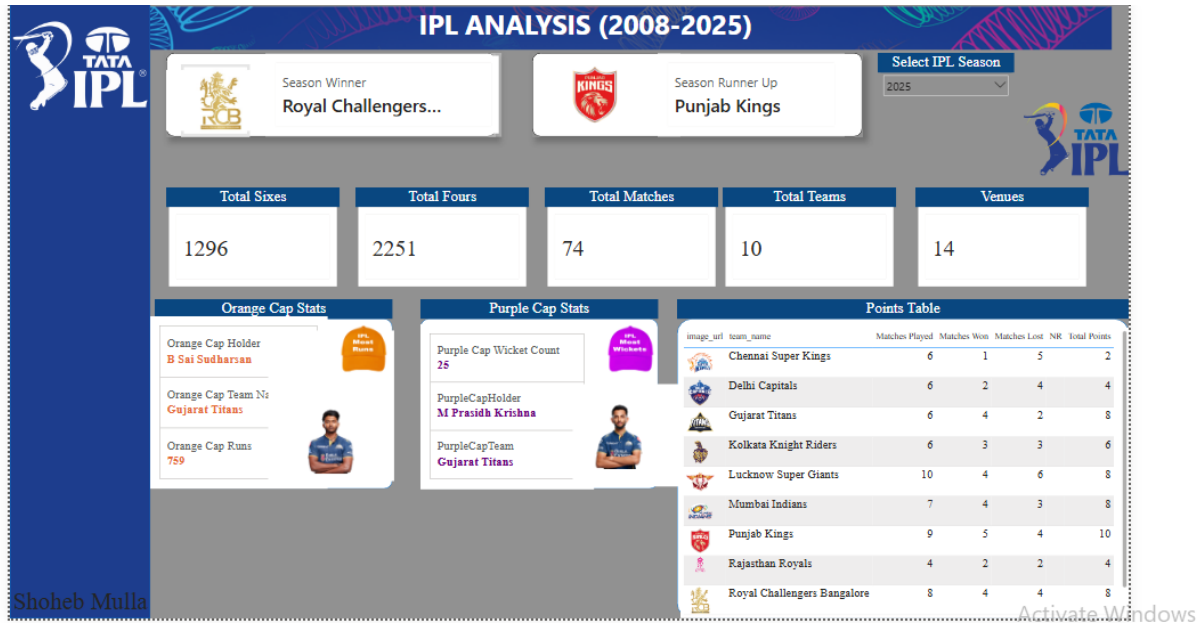


IPL ANALYSIS DASHBOARD

Project Documentation | 2008-2025



Project Type	Sports Data Analytics / Business Intelligence
Tool	Microsoft Power BI
Analysis Period	2008-2025
Author	Shoheb Mulla

This document describes the objective, dashboard design, analytical features, Power BI techniques, and intended use of the IPL Analysis Dashboard.

1. Project Overview

The IPL Analysis Dashboard is an interactive Power BI project created to analyze Indian Premier League data across seasons from 2008 to 2025. The dashboard combines key season, team, match, batting, bowling, and points-table information into a single interactive report.

Users can select an IPL season and view dynamically updated KPIs and performance information. The dashboard is designed as a portfolio-level Business Intelligence project demonstrating data preparation, DAX calculations, data modeling, and visual design.

2. Project Objectives

- Analyze IPL seasons from 2008 to 2025.
- Create an interactive season selector for dynamic analysis.
- Display important match and team KPIs.
- Identify Orange Cap and Purple Cap performers for a selected season.
- Present team performance through a points table.
- Demonstrate the use of DAX measures and Power BI visualizations.
- Convert raw cricket data into an easy-to-understand analytical dashboard.

3. Tools and Technologies

Technology	Purpose
Microsoft Power BI	Dashboard creation, data modeling and interactive visualization
Power Query	Data cleaning and transformation
DAX	Measures, calculations and dynamic season-based analysis
CSV / Tabular Data	Source data used for IPL analysis

4. Dashboard Components

Component	Description
Season Winner	Displays the winner of the selected IPL season.
Season Runner-up	Displays the team that finished as runner-up.
Total Sixes	Shows the total number of sixes for the selected season.
Total Fours	Shows the total number of fours for the selected season.
Total Matches	Shows the number of matches in the selected season.
Total Teams	Shows the number of teams participating in the selected season.
Venues	Shows the number of venues represented in the selected season.
Orange Cap Stats	Shows the leading run scorer, team and runs for the selected season.
Purple Cap Stats	Shows the leading wicket-taker, team and wickets for the selected season.
Points Table	Compares teams using matches played, wins, losses, no-results and points.

5. Data and Data Preparation

The dashboard is intended to use IPL match-level and ball-by-ball information. The match-level data supports season, team, venue, winner and result analysis, while ball-by-ball data supports detailed batting and bowling calculations such as fours, sixes, runs and wickets.

Typical preparation steps include:

- Checking column names and data types.
- Removing or handling duplicate records where required.
- Handling missing or blank values appropriately.
- Creating relationships between related tables.
- Creating calculated measures for dashboard KPIs.
- Filtering calculations according to the selected IPL season.

6. Data Modeling Approach

The Power BI model connects match information with ball-by-ball information and supporting team/player attributes. A relational model allows season filters to flow into the measures used by the dashboard.

A typical logical structure contains a match table at the match level and a ball-by-ball table at the delivery level, with common identifiers used to relate the two. Supporting team and player information can be used for cleaner reporting and filtering.

7. DAX and Analytical Calculations

DAX measures are used to make the dashboard dynamic. Functions such as SELECTEDVALUE, CALCULATE, FILTER, SUM, COUNTROWS, DISTINCTCOUNT, TOPN, RELATED, VAR and RETURN can be used to calculate season-specific KPIs and identify top performers.

Analysis	Purpose
Selected Season	Reads the season selected by the user.
Total Matches	Counts matches for the selected season.
Total Fours / Sixes	Aggregates boundary events for the selected season.
Matches Won / Lost	Calculates team results under the current filter context.
Total Points	Calculates team points from match outcomes.
Orange Cap	Identifies the highest run scorer for the selected season.
Purple Cap	Identifies the highest valid wicket-taker for the selected season.

8. Dashboard Design

The dashboard uses a cricket-themed visual layout with a clear hierarchy. The top section highlights the selected season's winner and runner-up. KPI cards provide a quick summary of the season, while the lower section focuses on individual player achievements and the team points table.

- Season slicer for interactive filtering.
- KPI cards for quick statistical summaries.

- Player cards for Orange Cap and Purple Cap information.
- Table visualization for team standings.
- Team and IPL branding elements for visual context.

9. Analytical Questions Answered

- Who won the selected IPL season?
- Who was the runner-up?
- How many matches were played?
- How many teams participated?
- How many venues were used?
- How many fours and sixes were scored?
- Who won the Orange Cap?
- Who won the Purple Cap?
- How many runs did the Orange Cap holder score?
- How many wickets did the Purple Cap holder take?
- How did teams perform in the points table?

10. Key Insights

The dashboard provides a convenient way to compare IPL seasons and examine team and player performance. Because the report is filter-driven, users can move from a high-level season summary to team standings and individual player achievements without changing the underlying report.

The dashboard is particularly useful for demonstrating how raw sports data can be transformed into business-style KPIs and interactive visual analytics.

11. Project Structure

Recommended GitHub repository structure:

```
IPL-Analysis-PowerBI/
├── README.md
├── Dashboard/
│   └── IPL_Analysis.pbix
├── Dataset/
│   ├── ipl_matches_data.csv
│   └── ball_by_ball_data.csv
├── Images/
│   └── IPL_Dashboard.png
├── Documentation/
└── IPL_Analysis_Project_Documentation.pdf
```

12. Skills Demonstrated

- Power BI dashboard development
- DAX measure creation
- Power Query data transformation
- Data modeling and relationships
- KPI development

- Interactive data visualization
- Sports analytics
- Analytical storytelling

13. Future Enhancements

- Add player-wise batting and bowling comparison pages.
- Add team-wise historical performance trends.
- Add win-percentage and net-run-rate analysis.
- Add venue-wise performance analysis.
- Add season-over-season comparison visuals.
- Add more interactive drill-through pages.
- Publish the dashboard to Power BI Service for online access, where appropriate.

14. Conclusion

The IPL Analysis Dashboard demonstrates an end-to-end Business Intelligence workflow: preparing structured cricket data, modeling relationships, creating DAX calculations, and presenting the results through an interactive Power BI dashboard. The project can serve as a portfolio project for demonstrating practical Data Analyst and Power BI skills.

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Data Analytics | Power BI | DAX | Python | SQL | Data Visualization